

5.1.1 Artificial Intelligence (AI)

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 02 of 12

★ Golden

5.1.1 Artificial Intelligence (AI) ★ Golden

LEARN IT

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

Understand the name in two parts:

- **Artificial** — it is *created by humans*; it is not natural.
- **Intelligence** — it can perform tasks that *require thinking*.

AI systems **use data to make decisions**. The more good data an AI system sees, the better it becomes at recognising patterns (repeating features) and predicting what to do.

Examples from the source:

1. **Face recognition** on a mobile phone to unlock the screen.
2. **YouTube suggesting videos** based on your interests.
3. **Chatbots** answering questions on a website (for example, a bank's help chat).

More Pakistani examples: Google Maps predicting traffic on Shahrah-e-Faisal, Urdu voice typing on your phone, banks flagging a suspicious card transaction, and AI tools that translate English into Urdu.

AI in smart systems

AI rarely works alone. It often acts as the **"brain"** of a system, working together with the **Internet of Things (IoT)** and **Data Analytics**:

- **IoT devices gather data** — they are the "eyes" (sensors).
- **Analytics sorts the data** — it organises and studies it.
- **AI makes the final decision** — it is the "brain".
- An **action** follows — something happens in the real world.

IoT (Eyes) — collects data



Analytics — sorts data



AI (Brain) — decides



Action

WORKED EXAMPLE

Smart air-conditioner in a hot Hyderabad room — trace the system:

Step 1 (IoT – eyes). A temperature sensor reads the room temperature: 36 °C.

Step 2 (Analytics). The readings of the last 10 minutes are organised: 34, 35, 35, 36 °C — the room is getting hotter.

Step 3 (AI – brain). The AI compares this with the comfortable limit (say 26 °C) and with the fact that a person is in the room (motion sensor). Decision: "Turn the AC on."

Step 4 (Action). A switch turns the AC on automatically.

No human pressed a button — the system sensed, thought and acted.

REMEMBER IT

Eyes – Sort – Brain – Do. IoT is the *eyes*, Analytics *sorts*, AI is the *brain*, and the actuator *does* the action. Think of a cricket fielder: his eyes see the ball (IoT), he judges speed and direction (analytics), decides to dive (AI), and his body moves (action).

COMMON MISTAKE

Mistake: "AI means robots." **Fix:** Most AI is just *software* — YouTube suggestions, spam filters and face unlock have no robot body. A robot may *use* AI, but AI itself is the thinking software.

اردو میں سمجھیں

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔ AI ڈیٹا کی مدد سے فیصلے کرتی ہے، جیسے چہرہ پہچان کر فون کھولنا یا یوٹیوب پر ویڈیوز تجویز کرنا۔ ایک اسمارٹ سسٹم میں IoT آنکھوں کا کام کرتا ہے اور AI دماغ کا۔

EXAM TIP

Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks).

To score full marks: (1) write the two-part meaning of the name — *artificial* = *made by humans*, *intelligence* = *can think/decide*; (2) say "AI uses data to make decisions"; (3) give at least two examples; (4) for the "together" question, draw the flow IoT → Analytics → AI → Action and explain it with the room-AC example.

CHECK YOURSELF

1. Why is AI called "artificial"?
2. In a smart system, which technology is the "brain" and which is the "eyes"?
3. Is a spam filter in Gmail an example of AI? Why?

Answers: 1. Because it is created by humans. 2. AI is the brain; IoT is the eyes. 3. Yes — it uses data (past spam emails) to decide whether a new email is spam, a task that needs "thinking".

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. AI is:

- a. Machines performing tasks that need human intelligence
- b. Any Excel sheet
- c. A copper cable
- d. A primary key

MCQ 2. Machine learning is AI that:

- a. Learns patterns from data
- b. Only follows a fixed 5-line script forever
- c. Is analog only
- d. Is a pie chart

MCQ 3. A chatbot is an:

- a. AI application
- b. OSI hub
- c. AND gate
- d. MVP canvas

MCQ 4. AI needs:

- a. Data + model + compute
- b. Only a mouse
- c. Only paper
- d. Only Agile stickers

MCQ 5. An exam risk of AI answers is:

- a. Hallucination / wrong confident text
- b. Too much NAND
- c. Too much OSI
- d. Too much bubble sort

MCQ 6. This is a ★ topic because papers ask:

- a. Definition + examples + one issue
- b. Only cabling colours
- c. Only Access wizards
- d. Only Big O proofs

Extra short questions

1. Define AI with two examples.
2. AI vs ordinary software.
3. One benefit and one risk in exams/education.
4. Name two Pakistani/education uses.

Extra long question

1. Explain AI to a Class XI student: definition, two components, two applications, two ethical issues. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Intelligence in machines. Face unlock; Google Translate; board-result chatbots.
2. Ordinary: fixed rules. AI: improves from data / handles messy input.
3. Benefit: tutoring. Risk: cheating / wrong answers.
4. Urdu OCR; adaptive quizzes; NADRA biometrics (accept reasonable).

Long

1. Def. Data+algorithms(+sensors). Apps: health imaging, spam filter. Ethics: bias, privacy, job loss. One sentence on human remaining responsible.